

	SOFTWARE REQUIREMENTS SPECIFICATION OF Spacecraft Status Display and Commanding Software 1.0	Doc. No: SCG-GRCD2-SW-2021-01 Version : 1.0
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SOFTWARE REQUIREMENTS SPECIFICATION

SPAcecraft **S**tatus **D**isplay **A**nd **C**ommanding **S**oftware (SPASDACS) 1.0

GEOSAT RF & Payload Checkout Division-2
Spacecraft Checkout Group

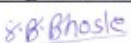
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UR RAO SATELLITE CENTRE
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BANGALORE

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AUTHENTICATION

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CHANGE HISTORY

Version Number	Date	Affected Section, Figure, Table	Nature of change [A,M,D] *	Description
1.0	02/08/2021	-	New	

* A – Addition, D – Deletion, M-Modification

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1. Introduction

1.1 Purpose

Spacecraft Status Display and Commanding Software is for Real Time display of an ‘Active Block diagram’ of a spacecraft subsystem for Pictorial display of Telemetry/SMON/ADC data Status.

This software gives pictorial representation of spacecraft telemetry instead of text display. Pictorial display are more intuitive then text.

“A Picture is worth a thousand words”

As shown in GUI prototype Fig.1 below,

This software shall enable user to draw system block diagram and allows changing its visual appearance based on live data (telemetry/SMON/ADC data).

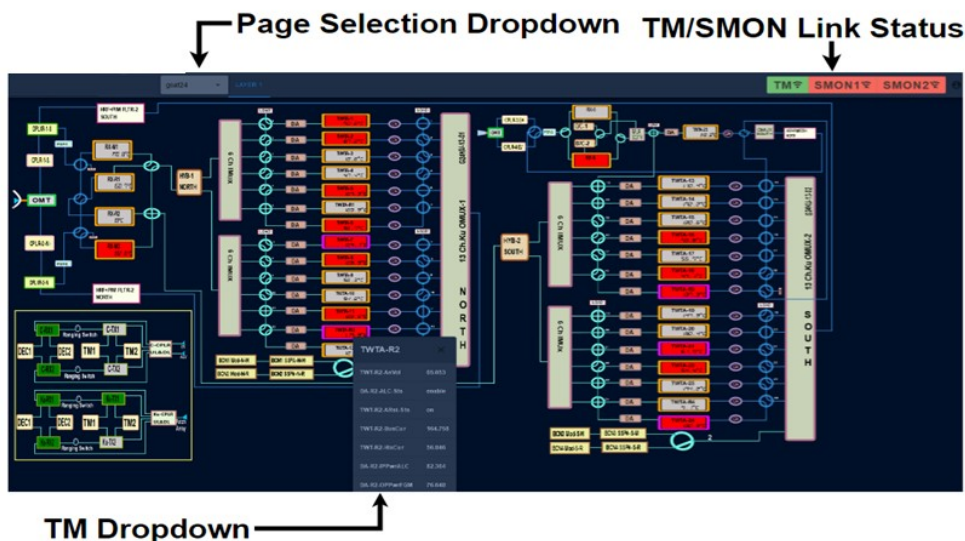


Figure 1: SPASDACS GUI prototype

1. It should provide alerts (audio & visual) on invalid / prohibitive combinations defined by the user.
2. Provision should be there to Conditional Hide/Display of a block.
3. Should indicate Test path/Signal flow path.
4. On Mouseover of block TM should be visible.
5. Provision should be there to select different pages
6. Provision shall be there to indicate telemetry/SMON/ADC link status.
7. Built in Simulator should be there to test the created block diagram.

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1.2 Scope

This document outlines the requirements of the **Spacecraft Status Display and Commanding Software (SPASDACS)** product which is proposed to be developed. This document is an outcome of requirements generated by existing software, experience gained during the Testing of various satellites and after discussions and feedback from the user teams.



1.3 Definitions, Acronyms and Abbreviations

S. No	Acronym / Abbreviation	Definition
1.	SPASDACS	Spacecraft status display and commanding software
2.	PASDACS	Payload status display and commanding software.
3.	SCC	Spacecraft Checkout Computer
4.	PCC	Payload Checkout Computer
5.	TM	Telemetry
6.	TC	Telecommand
7.	GUI	Graphical User Interface
8.	SCOS	Special Checkout System
9.	SMON	SCOS Monitoring
10.	ADC	Analog to Digital Converter

Table 1: Acronyms & Abbreviations

1.4 References

Document Name	Document Number	Publishing Date
PASDACS REQUIREMENT DOCUMENT	Not Available	--

Table 2: References

1.5 Overview of the document

This document describes the functional and interface requirements of SPASDACS 1.0
Section 2 gives the overall description of the Software
Section 3 gives the detailed description of functional and interface requirements
Section 4 lists the requirement traceability matrix



2. Overall Description

This software product will be used for visual representation of spacecraft subsystems. This software shall be used during IST at checkout.

2.1 Product Perspective

The application shall be implemented to run on SCC or PCC or any other PC which is on CHECKNET with windows/Linux operating system. The proposed software is required to show the 'Active Block schematic of the spacecraft systems' of the selected spacecraft on the computer screen.

2.2 Product Functions

The major functional requirements of this software can be grouped into preparatory requirements and real time display requirements:

2.2.1 Preparatory requirements:

These requirements will address the preparation of block schematic of spacecraft systems, linking telemetry with schematic and creating user accounts.

1. Shall be able to create block schematics of systems with basic shapes rectangle, circle, ellipse, line, and triangle and the shape attributes color, size & visibility shall change as per telemetry values assigned to that shape.
2. Shall have a provision to assign telemetry to block schematic.
3. Shall have provision to create users and assign roles (as defined below) to them.
 - a) Admin: Shall be able create/edit schematics, assign/change telemetry association with schematic.
 - b) Guest: Shall be able to view the schematic at run-time.

2.2.2 Real-time display requirements:

These requirements will address functionality of the system in execution environment where spacecraft live data will be received and displayed visually to the user.

4. Shall be able to receive live telemetry, SMON, ADC data from selected SCC Server.
5. Shall have provision to switch between multiple block schematics.
6. Shall have provision to zoom in zoom out.
7. Mouseover on any subsystem block shall display telemetry dropdown of assigned telemetry.
8. Shall provide audio and visual alert on limit crossing of all telemetry parameters used in the schematic.
9. Status change shall be accompanied by audio & blinking (blinking duration is 1 second) of that particular block.
10. Shall provide audio and visual alert on Data break from SCC.
11. Provision shall be there to switch to Preparatory mode any time.

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2.3 Constraints

SPASDACS software has to receive data from SCC server via web socket protocol. So this software should comply with protocol requirements for data reception.

2.4 Assumptions and Dependencies

1. TM, SMON and ADC data publishers should be running in SCC.

2.5 Apportioning of requirements

The following requirements are planned for the next version of the software:

- Commanding feature to send commands to spacecraft and checkout instruments from visual display.



3. Specific Requirements

3.1 External Interface Requirements

3.1.1 Spacecraft/Payload Checkout Computer

TM, SMON and ADC data publishers should be running In Spacecraft Checkout Computer.

3.1.3 Software Interfaces

Interface with Telemetry applications

The software shall connect to the telemetry/smon/adc data publisher software which will be running on SCC. The data format and subscription procedure is as follows

The publisher software will publish data using web socket protocol. The user has to subscribe to required data by sending subscription request.

Telemetry Data Interface:

To subscribe/unsubscribe to telemetry the following request to be sent to web socket server

```
{"action": "subscribe/unsubscribe", "params": ["Frame-id", "OBT"] }
```

After subscription the telemetry is transmitted continuously to client based on sampling rate of the telemetry parameter.

TM Data message format:

```
{"param_id":"","param":"","source_info":"","raw_count":"","proc_value":"","time_stamp":"","upper_limit":"","lower_limit":"","err_desc":""}
```

Telecommand Data Interface:

To subscribe/unsubscribe to telecommand the following request to be sent to web socket server

```
{"action": "subscribe/unsubscribe"}
```

After subscription, whenever the telecommand is issued to spacecraft the information is transmitted to client in the following json format

```
{"cmd":"","code":"","full_code":"","data_part":"status":"","time":""}
```

SMON/ADC Data Interface:



To subscribe/unsubscribe to SMON/ADC data the following request to be sent to web socket server

```
{"action": "subscribe/unsubscribe"}
```

After subscription, every 4 seconds the information is transmitted to client in the following json format

```
{" paramlist": [{ "param": "", "value": "" }, { "param": "", "value": "" } ... ],  
"stream": "", "seqcount": "", "time": "", "error": "" }
```

The Software interface of SPASDACS should cater to these requirements to receive data.

3.2 Graphical User Interface

Graphical user interface shall exist for following operation scenarios:

1. Block schematic preparation.
2. Telemetry association with block schematic
3. Real time display

3.2.1 Block schematic preparation

Graphical User interface shall exist to create/edit System block diagram as per user requirement. It should have following features

- a) Shall be able to draw different shapes (rectangle, circle, ellipse, line, and triangle).
- b) Shall be able to scale the shapes.
- c) Shall have cut/paste/delete/align/group options for shapes.
- d) Shall have option to change size/color/border color/border-width of shapes.
- e) Shall have option to save/open/edit schematic as a file.
- f) Shall have option to import images onto the editor.
- g) Shall have provision to import and use custom shapes (other than basic shapes (point a)).

3.2.2 Telemetry/SMON/ADC data association with block schematic

1. Graphical user interface shall exist with provision to associate telemetry/smon/adc data with system block schematic.
2. Graphical user interface shall exist to edit telemetry/smon/adc data association with block schematic.

3.2.3 Real Time display

1. Graphical user interface shall have provision to switch between multiple block schematics.
 2. Graphical user interface shall display status change accompanied by audio & blinking (blinking duration is 1 second) of that particular block.
 3. Graphical user interface shall display telemetry associated with block as dropdown on Mouseover of that element block.
 4. Graphical user interface shall display telemetry/smon/adc data link status.
- Sample GUI diagram is shown below



Functional Requirements

3.2.4 Creating system block /representational diagram.

Application shall have utility to create block/representational diagram of the spacecraft subsystem. The utility shall have following features

1. Easy to use.
2. Shall be able to draw different shapes (rectangle, circle, ellipse, line, and triangle).
3. Provision to create custom sub systems blocks library for ease of use.
4. Shall have all features of a standard editor like save, copy, paste, redo, undo etc.

3.2.5 Linking Telemetry/SMON/ADC data with block attributes.

Provision shall be there for each element in block diagram to change it visual attributes based on telemetry/SMON/ADC data. The following visual attributes shall be programmable based on telemetry.

1. Color.
2. Border color
3. Width.
4. Height.
5. Rotation angle.
6. Position(x, y)
7. Hide & show.

3.2.6 Alerts on Telemetry/SMON/ADC Data Break

Provision shall be there to alert (Visual & Audio) user when there is a telemetry/SMON/ADC data break.

3.2.7 Invalid condition/status of the blocks

Provision shall be there to alert (Visual & Audio) when telemetry of a system becomes invalid. User shall be able to define Invalid conditions.

3.2.8 System Status Change Indication

System digital state change shall be visually indicated by means of blinking (for 1 second).

3.2.9 Telemetry Dropdown

On mouse over, the block diagram shall display telemetry data list defined by the user. On each block user will define telemetry mnemonics those mnemonics with data



shall be displayed on mouse over. The drop down shall update values on real time based on telemetry changes.

3.2.10 Multiple display pages

Provision shall exist to add multiple display pages. The block schematic if split over more than one page, shall have links to switch back & forth between pages.

3.2.11 The display shall be viewable on any PC over a LAN

Provision shall be there to view visual display on any PC in the network without any additional installations.

3.2.12 Telemetry/SMON/ADC data Simulator

Telemetry simulation software shall be made available to test the display.

3.3. Performance Requirements

- This package shall be able to run on a PC with RAM of 2 GB.

3.4. Design Constraints

1. The software interfaces for receiving TM/TC/SMON/ADC data should be web socket based as the current interface at SCC for these data streams is through web sockets.
2. The Software should run on windows/Linux OS.

3.5. Software System attributes

Reliability Software reliability is the probability of failure free software operation for a specified period of time in a specified environment. To increase the reliability aspects, this software shall undergo proper verification & validation with exhaustive test cases to cover all possibilities of error handling features

Availability Proper and systematic back-up and auto restart features to be planned to increase the availability of the software

Maintainability The software shall follow modular design, loose coupling between modules and data driven to cater to easy maintenance across different projects and varying requirements in the future

SPASDACS Software Goals:

1. The displays shall be aesthetically pleasing.



2. At a glance user shall be able to detect any issues in the telemetry and shall be able to detect state of the spacecraft.
3. Spacecraft Safety shall be ensured throughout the process of commanding from the display.
4. Shall be user-friendly



4. Requirements Traceability Matrix

Software Requirement Id	Section	Description
RID 1	3.2.1	Utility to create system block/representational diagram
RID 2	3.2.2	GUI to associate TM/SMON/ADC data.
RID 3	3.2.3	GUI to display TM/SMON/ADC real time data status
RID 4	3.2.4	Provision to create block diagram library
RID 5	3.2.5	Linking TM/SMON/ADC data to Visual attribute of blocks
RID 6	3.2.6	Alerts on TM/SMON/ADC Data Break
RID 7	3.2.7	Invalid condition/status of the blocks
RID 8	3.2.8	System Status Change Indication
RID 9	3.2.9	Telemetry Dropdown
RID 10	3.2.10	Multiple display pages
RID 11	3.2.11	The display shall be viewable on any PC over a LAN
RID 12	3.2.12	Telemetry/SMON/ADC data Simulator

Table 3: Requirements Traceability